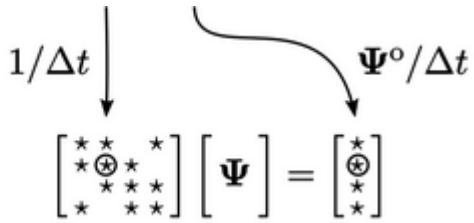
Notes on Computational Fluid Dynamics: General Principles

3.17 Time discretisation

A local time derivative $\partial\Psi/\partial t$ in an equation can be discretised as a finite difference in time. Time t is expressed in discrete intervals, or *steps*, of duration Δt.

The **Euler** scheme calculates the derivative from the field at the current time Ψ and the previous, or *old*, time level Ψ^o by:

$$\frac{\partial\Psi}{\partial t} \rightarrow \frac{\Psi - \Psi^o}{\Delta t} \tag{3.21}$$



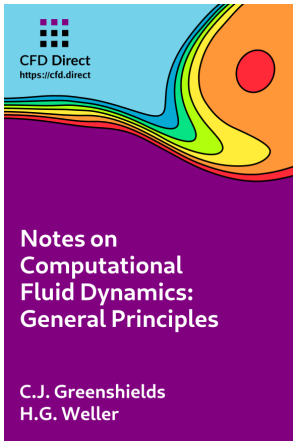
Time discretisation thereby contributes to the *diagonal* (*only*!) coefficients of the matrix $[\mathbf{A}]$ and to the source vector $[\mathbf{b}]$.

Courant number

For a 1D domain in the x-direction, the Courant number is the following dimensionless parameter for each cell of length Δx:

$$\mathrm{Co} = \frac{u_x \Delta t}{\Delta x}. \tag{3.22}$$

The Courant number originates from the solution of 1D advection, *e.g.* Eq. (2.32), with the Euler time scheme of Eq. (3.21) and an *explicit* upwind advection scheme from Sec. 3.10. For that case, the Courant-Friedrichs-Lewy (CFL) condition[12] for convergence is $\mathrm{Co} \leq 1$ across *all cells*. $\mathrm{Co} = 1$ corresponds to a fluid particle moving across one cell in one



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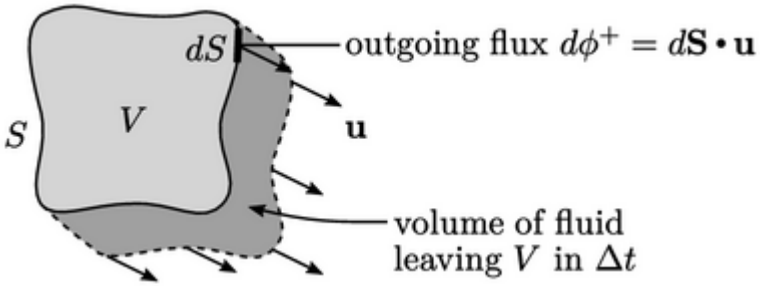
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time step, so its relevance to solution convergence is perhaps unsurprising.

In an *explicit* solution, the convergence limit can reduce further to $\mathrm{Co} \leq 0.5$ or $\mathrm{Co} \leq 0.25$, with more accurate schemes for advection. But the finite volume method is generally *implicit* so stability can then be maintained with a higher maximum value of Co. Temporal accuracy is then the important consideration when choosing the Co of a simulation, both in terms of its mean and maximum value across all cells in the domain.



It is therefore important to monitor Courant number which needs to be calculated for 3D problems. Explicit, Euler, upwind discretisation of advection can be presented in 3D as follows:

$$\Psi = \Psi^{\mathrm{o}} - \frac{\Delta t}{V} \sum_{\mathrm{f}} \phi^{+} \Psi^{\mathrm{o}} - \frac{\Delta t}{V} \sum_{\mathrm{f}} \phi^{-} \Psi^{\mathrm{o}}_{\mathrm{N}} = 0. \tag{3.23}$$

Here ϕ^{+} are positive fluxes which transports Ψ^{o} out of the cell of interest and ϕ^{-} are negative fluxes which transports $\Psi^{\mathrm{o}}_{\mathrm{N}}$ from neighbouring cells. Since the CFL condition, $\mathrm{Co} \leq 1$, requires that the coefficient of Ψ^{o} cannot be negative, it follows that

$$\mathrm{Co} = \frac{\Delta t}{V} \sum_{\mathrm{f}} \phi^{+}. \tag{3.24}$$

This 3D Co represents the volume of fluid leaving the cell in one time step, as a fraction of the cell volume, as shown above.

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[12] after Richard Courant, Kurt Friedrichs and Hans Lewy, *Uber die partiellen Differenzengleichungen der mathematischen Physik*, 1928.

Chris Greenshields Henry Weller 11th April 2022 Notes on CFD

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